IT Employee Lifecycle Automation System

System Requirements Documentation

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Table of Contents

1. Customer Problem Statement
2. Glossary of Terms
3. System Requirements
   - 3.1. Functional Requirements
   - 3.2. Nonfunctional Requirements (FURPS)
4. Functional Requirements Specification
   - 4.1. Stakeholders
   - 4.2. Actors and Goals
   - 4.3. Use Cases
   - 4.4. Class Diagram — Component Overview
   - 4.5. Enumerations
   - 4.6. Data Types and Operation Signatures
   - 4.7. Class Relationships
5. System Sequence Diagrams
6. Activity Diagram
7. User Interface Specification
   - 7.1. Global Layout and Navigation
   - 7.2. UI-1: Main Dashboard
   - 7.3. UI-2: Department Mapping Configuration
   - 7.4. UI-3: Task Detail View
   - 7.5. UI-4: Audit Log Viewer
   - 7.6. UI-5: Report Generation
   - 7.7. UI-6: Employee Search & Status Lookup
   - 7.8. User Effort Estimation
8. Project Plan

9. References

1. Customer Problem Statement

Organizations of all sizes rely on IT departments to provision and deprovision user accounts throughout the employee lifecycle. When a new employee joins, IT administrators must create Active Directory accounts, assign appropriate security groups based on department and role, and grant access to necessary resources. This process typically begins when IT receives notification via email or helpdesk ticket — often the day of or after the employee starts — resulting in delayed access and lost productivity during critical onboarding periods.

The offboarding process presents even greater challenges. When employees depart, IT frequently receives late or incomplete notifications, leaving former employees with active credentials and access to sensitive company resources. These orphaned accounts create significant security vulnerabilities and compliance risks. Additionally, manual provisioning leads to inconsistency: different IT staff may assign different permissions to employees in identical roles, creating access disparities and audit failures.

An automated system is needed that continuously monitors the HR system for personnel changes and automatically executes corresponding account actions — creating accounts for new hires with role-appropriate permissions and promptly disabling accounts for departing employees. The system provides a centralized dashboard for oversight, manual intervention capabilities, and comprehensive audit logging to ensure compliance.

2. Glossary of Terms

Term	Definition
Employee Lifecycle	The stages an employee progresses through within an organization: onboarding (joining), active employment, role changes, and offboarding (departure).
Provisioning	The process of creating user accounts and assigning appropriate access permissions when an employee joins the organization.
Deprovisioning	The process of disabling or removing user accounts and revoking access when an employee leaves the organization.
Orphaned Account	A user account that remains active after an employee has left the organization, creating a security vulnerability.
Role-Permission Mapping	A configuration that defines which security groups and access levels are automatically assigned based on an employee's department and job title.

Term	Definition
Polling Interval	The frequency at which the system checks the HR database for changes in employee status.
Audit Log	A chronological record of all account lifecycle events, including who performed actions and when.
Distinguished Name (DN)	The full LDAP path identifying a user or group object within Active Directory (e.g., CN=jsmith,OU=ITLifecycle,DC=stechlab,DC=net).
LDAP	Lightweight Directory Access Protocol — the protocol used to communicate with Active Directory.
OrangeHRM	The open-source Human Resource Management system used as the authoritative source of employee data.

3. System Requirements

3.1 Functional Requirements

No.	Priority	Description
REQ-1	High	The system shall automatically detect new employees added to the HR management system.
REQ-2	High	The system shall automatically create Active Directory accounts for new employees.
REQ-3	High	The system shall assign security groups based on configured department and role mappings.
REQ-4	High	The system shall automatically detect employee terminations in the HR management system.
REQ-5	High	The system shall automatically disable Active Directory accounts for terminated employees.
REQ-6	High	IT administrators shall be able to configure mappings between departments/roles and Active Directory security groups.
REQ-7	High	IT administrators shall be able to manually approve, modify, or override automated account actions.
REQ-8	Medium	IT administrators shall be able to configure the polling interval for HR system synchronization.
REQ-9	High	The system shall maintain comprehensive audit logs of all account lifecycle events.
REQ-10	Medium	IT administrators shall be able to generate compliance reports for specified date ranges.
REQ-11	Medium	HR personnel shall be able to view the provisioning status of newly entered employees.
REQ-12	Low	HR personnel shall be able to flag high-priority onboarding requests for expedited processing.
REQ-13	Medium	Department managers shall be able to view account status for their direct reports.
REQ-14	Low	Department managers shall be able to submit requests for additional access or permissions for team members.

No.	Priority	Description
REQ-15	Medium	The system shall send notifications to stakeholders when account actions are completed or require attention.

3.2 Nonfunctional Requirements (FURPS)

No.	Category	Priority	Description
NFR-1	Functionality	High	The system shall integrate with OrangeHRM via direct MySQL database polling.
NFR-2	Functionality	High	The system shall communicate with Active Directory via LDAP or LDAPS protocol.
NFR-3	Usability	Medium	The dashboard shall display pending, completed, and failed tasks in a clear visual layout with status indicators.
NFR-4	Usability	Medium	The system shall provide intuitive navigation between different functional areas via a persistent sidebar.
NFR-5	Reliability	High	The system shall retry failed operations up to a configurable limit (default: 3) before marking them as failed.
NFR-6	Reliability	High	The system shall continue operating normally if individual account operations fail (graceful degradation).
NFR-7	Performance	Medium	The system shall detect and process new employee changes within 5 minutes of the configured polling interval.
NFR-8	Performance	Medium	Account creation and disabling operations shall complete within 30 seconds.
NFR-9	Supportability	Medium	The system shall log all errors with sufficient detail (timestamps, error codes, context) for troubleshooting.
NFR-10	Supportability	Low	The system shall support configuration changes (mappings, intervals, notifications) without code modifications.

4. Functional Requirements Specification

4.1 Stakeholders

Stakeholder	Interest in the System
IT Administrators	Primary users who manage account lifecycle operations, configure automation rules, handle exceptions, and ensure security compliance. They benefit from reduced manual workload and fewer errors.
HR Personnel	Staff who maintain employee records in OrangeHRM and need visibility into IT provisioning status to confirm that new hires receive timely access and departing employees are offboarded.
Department Managers	Leaders responsible for ensuring their new hires have appropriate access on day one and that departing team members are properly offboarded. They may request additional permissions for team members.
Compliance Officers	Reviewers who require audit trails to verify that account access changes follow organizational policies and regulatory requirements.
CIO / IT Director	Project sponsor who seeks reduced operational cost, improved security posture, and elimination of orphaned accounts through automation.
Employees (End Users)	Indirect beneficiaries who receive timely account access upon hiring, ensuring productive first days without waiting for manual IT provisioning.

4.2 Actors and Goals

Primary Actors

- **IT Administrator:** Configures automation rules, manages department-to-security-group mappings, approves or overrides automated account actions, views audit logs, generates compliance reports, and searches employee account status.
- **HR Personnel:** Views the provisioning status of newly entered employees, flags high-priority onboarding requests for expedited processing, confirms successful completion of offboarding procedures, and accesses a read-only view of employee account status.
- **Department Manager:** Views account status for their direct reports, submits requests for additional access or permissions for team members, and receives notifications when team member accounts are provisioned or modified.

Secondary Actors

- **System (Automation Engine):** Polls the OrangeHRM database at configured intervals, automatically creates or disables Active Directory accounts, assigns security groups based on role-permission mappings, sends notifications, and retries failed operations.
- **OrangeHRM (External System):** The human resource management system that serves as the authoritative source of employee data.
- **Active Directory (External System):** The directory service that receives account creation, modification, and disabling commands via LDAP protocol.

4.3 Use Cases

IT Administrator (total: 25 points)

Use Case	Description	Est.
Login/Logout	Authenticate to the system using credentials and manage session with JWT-based access control	4
View Dashboard	View a real-time dashboard displaying pending, in-progress, completed, and failed onboarding/offboarding tasks with statistics	2
Configure Role-Permission Mappings	Set up and maintain mappings between HR departments/job roles and Active Directory security groups	4
Approve/Override Account Actions	Review pending automated account actions and manually approve, modify, or reject them before execution	4
Configure Polling Interval	Set the frequency at which the system polls OrangeHRM for employee status changes	2
View Audit Logs	Browse and search a chronological record of all account lifecycle events for compliance and troubleshooting	3
Generate Compliance Reports	Create and export compliance or activity summary reports for specified date ranges and department filters	4
Search Employee Status	Look up any employee by name, ID, or email and view their current Active Directory account status	2

HR Personnel (total: 8 points)

Use Case	Description	Est.
View Provisioning Status	Check the current provisioning status of newly entered employees to confirm account creation is underway	2
Flag High-Priority Onboarding	Mark an onboarding request as urgent so the system processes it ahead of normal-priority tasks	2

Use Case	Description	Est.
Confirm Offboarding	Verify and acknowledge that offboarding procedures have been successfully completed for a departed employee	2
View Employee Account Status	Access a read-only view of any employee's current Active Directory account status	2

Department Manager (total: 7 points)

Use Case	Description	Est.
View Team Account Status	View the current account provisioning status for all direct reports in the department	2
Request Additional Access	Submit a request for additional security groups or permissions for a team member, routed to IT for approval	3
Receive Notifications	Receive alerts when team member accounts are provisioned, modified, or when actions require attention	2

System — Automation Engine (total: 18 points)

Use Case	Description	Est.
Poll HRM for Changes	Periodically query the OrangeHRM database to detect new hires and terminations	3
Process Onboarding	Create a new AD account for a detected new hire and trigger security group assignment and notifications	4
Process Offboarding	Disable the AD account and revoke group memberships for a detected terminated employee	3
Assign Security Groups	Automatically assign AD security groups based on the employee's department and role mappings	3
Send Notifications	Dispatch in-app notifications to relevant stakeholders when account actions complete or fail	3
Retry Failed Operations	Automatically retry a failed account operation up to the configured limit before marking the task as failed	2

Grand Total: 58 points (~29 engineer-days)

Use Case Relationships

<<include>> relationships (mandatory behavior):

- Process Onboarding **<<include>>** Assign Security Groups — Every onboarding task always assigns security groups based on department/role mappings.

- Process Onboarding **<<include>>** Send Notifications — Stakeholders are always notified after an onboarding action completes or fails.
- Process Offboarding **<<include>>** Send Notifications — Stakeholders are always notified after an offboarding action completes or fails.

<<extend>> relationships (optional/conditional behavior):

- Approve/Override Account Actions **<<extend>>** Process Onboarding — An IT administrator may optionally intervene before onboarding executes.
- Approve/Override Account Actions **<<extend>>** Process Offboarding — An IT administrator may optionally intervene before offboarding executes.
- Flag High-Priority Onboarding **<<extend>>** Process Onboarding — HR personnel may optionally flag a task for expedited processing.
- Retry Failed Operations **<<extend>>** Process Onboarding — The system retries only if the onboarding operation fails.
- Retry Failed Operations **<<extend>>** Process Offboarding — The system retries only if the offboarding operation fails.

4.4 Class Diagram — Component Overview

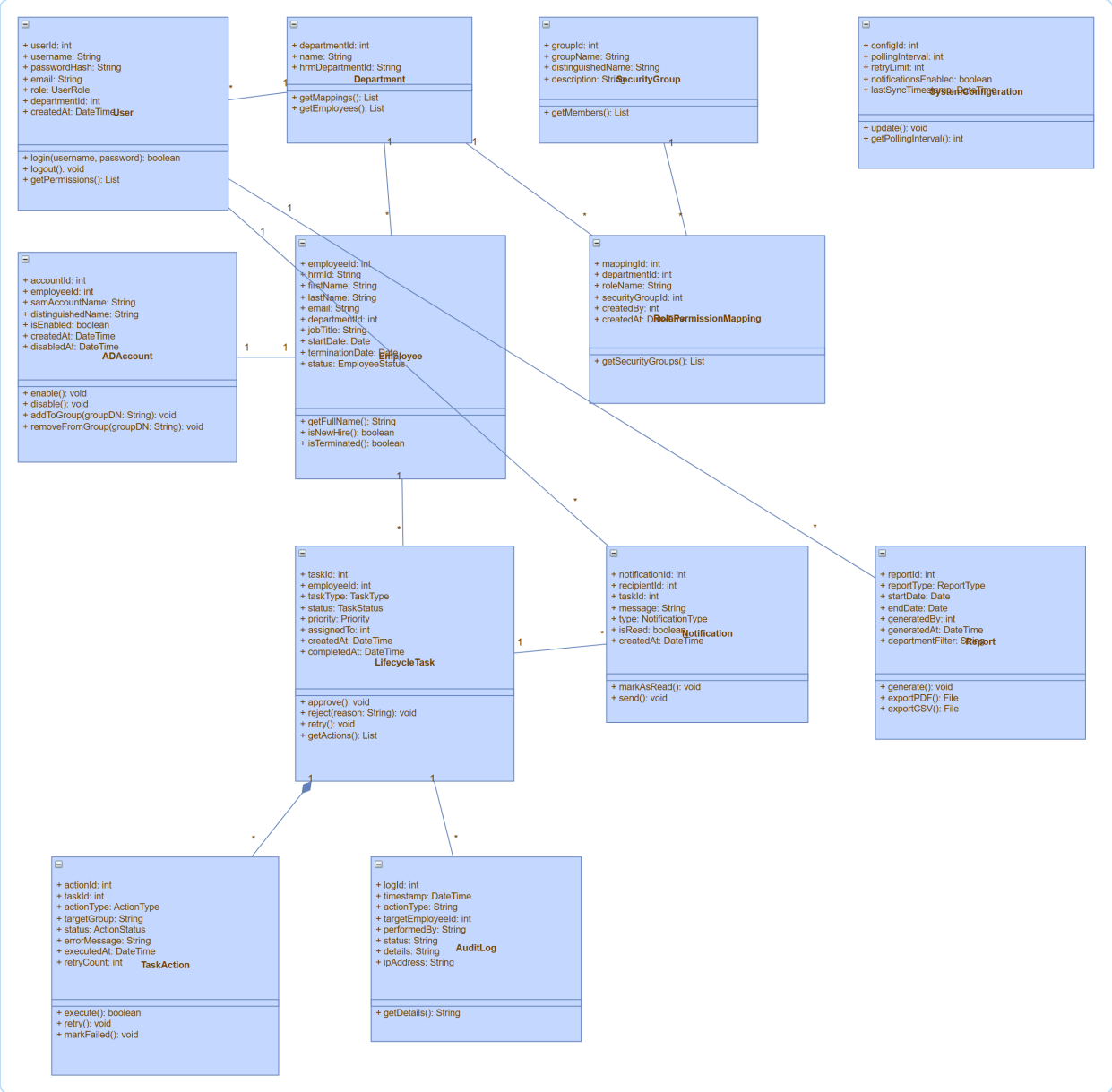


Figure: Class Diagram — 13 classes covering core entities, configuration, workflow, audit, and access control

Use Case Diagram

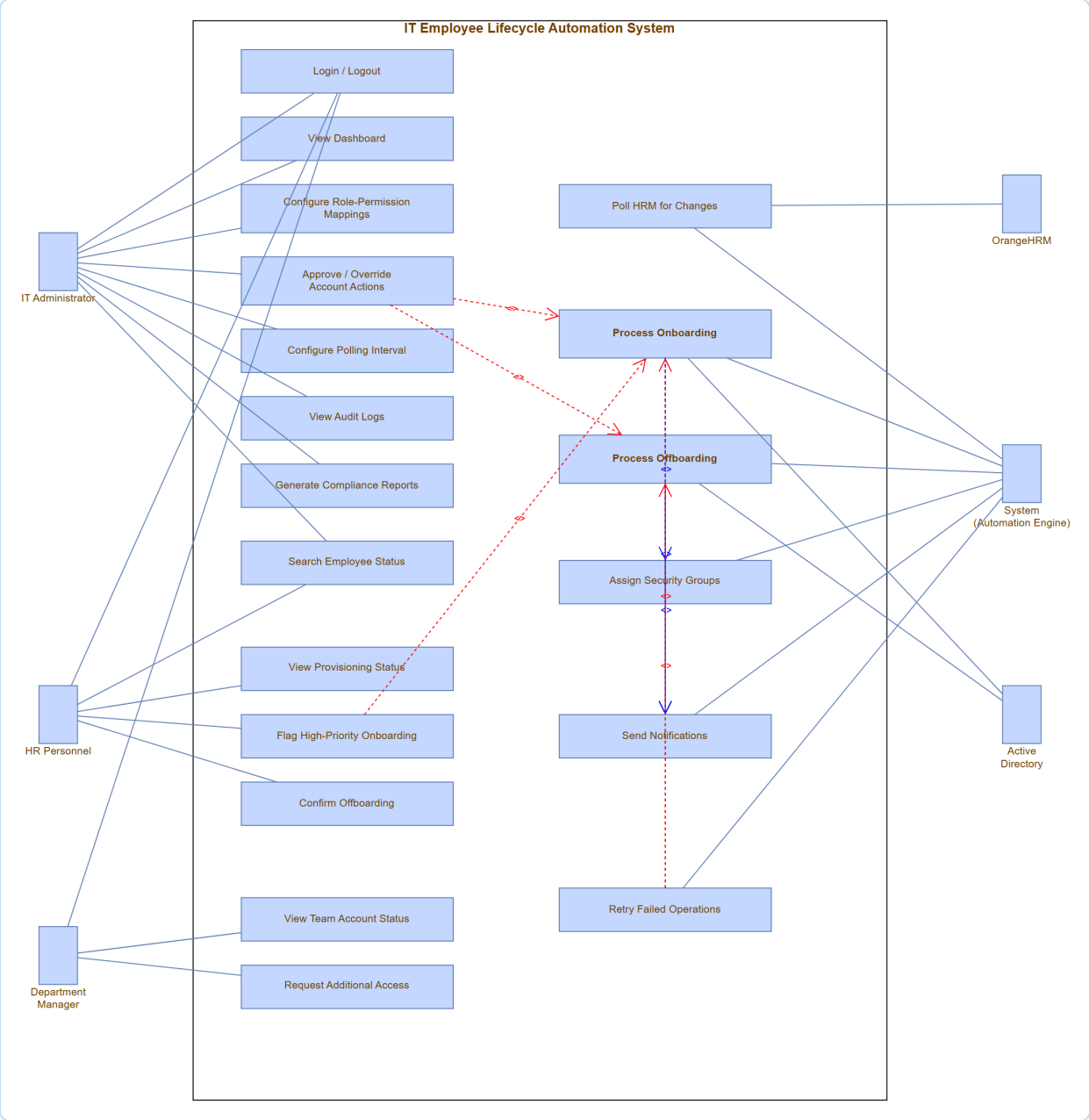


Figure: Use Case Diagram — all actors and use cases for the IT Employee Lifecycle Automation System

Category	Classes
Core Entities	Employee, ADAccount, Department, SecurityGroup
Configuration	RolePermissionMapping, SystemConfiguration
Workflow	LifecycleTask, TaskAction
Audit and Reporting	AuditLog, Notification, Report
Access Control	User

4.5 Enumerations

Enumeration	Values	Description
UserRole	IT_ADMIN, HR_PERSONNEL, DEPT_MANAGER, COMPLIANCE_OFFICER	Defines the role-based access level of a system user
EmployeeStatus	ACTIVE, TERMINATED, PENDING	Represents the current lifecycle state of an employee
TaskType	ONBOARD, OFFBOARD	Categorizes a lifecycle task as either onboarding or offboarding
TaskStatus	PENDING, IN_PROGRESS, COMPLETED, FAILED	Tracks the execution state of a lifecycle task
ActionType	CREATE_ACCOUNT, DISABLE_ACCOUNT, ASSIGN_GROUP, REMOVE_GROUP	Specifies the type of Active Directory operation
ActionStatus	PENDING, EXECUTING, SUCCESS, FAILED	Tracks the execution state of an individual task action
Priority	NORMAL, HIGH	Indicates the processing priority of a lifecycle task
NotificationType	INFO, WARNING, ERROR	Categorizes the severity of a notification
ReportType	COMPLIANCE, ACTIVITY_SUMMARY, FAILED_ACTIONS	Specifies the type of report to generate

4.6 Data Types and Operation Signatures

Class: User

A system user who accesses the application. Each user has a role that determines their permissions.

Attribute	Type	Description
userId	int	Unique identifier for the user
username	String	Login username for authentication
passwordHash	String	Bcrypt-hashed password
email	String	Email address for notifications
role	UserRole	The user's role determining access level
departmentId	int	Foreign key referencing the user's department
createdAt	DateTime	Timestamp when the user account was created

Operation	Signature	Description
login	login(username: String, password: String): boolean	Authenticates credentials; returns true on success
logout	logout(): void	Terminates the user's active session
getPermissions	getPermissions(): List	Returns the list of permissions for the user's role

Class: Employee

An employee record synchronized from OrangeHRM. Represents a person whose account lifecycle is managed by this system.

Attribute	Type	Description
employeeId	int	Unique identifier in this system
hrmId	String	The employee's unique identifier in OrangeHRM
firstName	String	Employee's first name
lastName	String	Employee's last name
email	String	Employee's work email address
departmentId	int	Foreign key referencing the employee's department
jobTitle	String	The employee's job title/role
startDate	Date	The employee's hire date
terminationDate	Date	The employee's termination date (null if active)
status	EmployeeStatus	Current lifecycle state: ACTIVE, TERMINATED, or PENDING

Operation	Signature	Description
getFullName	getFullName(): String	Returns the concatenation of firstName and lastName
isNewHire	isNewHire(): boolean	Returns true if the employee was added within the current polling cycle
isTerminated	isTerminated(): boolean	Returns true if the employee's status is TERMINATED

Class: Department

Attribute	Type	Description
departmentId	int	Unique identifier for the department

Attribute	Type	Description
name	String	Display name of the department
hrmDepartmentId	String	The department's unique identifier in OrangeHRM

Operation	Signature	Description
getMappings	getMappings(): List	Returns all role-permission mappings for this department
getEmployees	getEmployees(): List	Returns all employees belonging to this department

Class: SecurityGroup

An Active Directory security group that controls access to resources.

Attribute	Type	Description
groupId	int	Unique identifier for the security group
groupName	String	Display name of the AD security group
distinguishedName	String	The full LDAP distinguished name (DN) of the group in Active Directory
description	String	A brief description of the group's purpose

Class: RolePermissionMapping

Maps a department and job role combination to an AD security group.

Attribute	Type	Description
mappingId	int	Unique identifier for the mapping
departmentId	int	Foreign key referencing the department
roleName	String	The job title/role this mapping applies to
securityGroupId	int	Foreign key referencing the target security group
createdBy	int	Foreign key referencing the user who created this mapping
createdAt	DateTime	Timestamp when the mapping was created

Class: LifecycleTask

A workflow task representing an onboarding or offboarding operation for a specific employee.

Attribute	Type	Description
taskId	int	Unique identifier for the task

Attribute	Type	Description
employeeId	int	Foreign key referencing the target employee
taskType	TaskType	ONBOARD or OFFBOARD
status	TaskStatus	PENDING, IN_PROGRESS, COMPLETED, or FAILED
priority	Priority	NORMAL or HIGH
assignedTo	int	Foreign key referencing the assigned user (null if automated)
createdAt	DateTime	Timestamp when the task was created
completedAt	DateTime	Timestamp when the task was completed

Operation	Signature	Description
approve	approve(): void	Marks the task as approved and triggers execution
reject	reject(reason: String): void	Cancels the task with a documented reason
retry	retry(): void	Resets failed actions and re-executes the task
getActions	getActions(): List	Returns all actions belonging to this task

Class: TaskAction

An individual atomic action within a lifecycle task.

Attribute	Type	Description
actionId	int	Unique identifier for the action
taskId	int	Foreign key referencing the parent task
actionType	ActionType	CREATE_ACCOUNT, DISABLE_ACCOUNT, ASSIGN_GROUP, or REMOVE_GROUP
targetGroup	String	The DN of the target security group (null for account actions)
status	ActionStatus	PENDING, EXECUTING, SUCCESS, or FAILED
errorMessage	String	Error details if the action failed
executedAt	DateTime	Timestamp when the action was last executed
retryCount	int	Number of retry attempts made

Operation	Signature	Description
execute	execute(): boolean	Performs the AD operation; returns true on success
retry	retry(): void	Increments retryCount and re-executes the action

Operation	Signature	Description
markFailed	markFailed(): void	Sets status to FAILED after exhausting retry attempts

Class: AuditLog

A chronological, immutable record of an account lifecycle event.

Attribute	Type	Description
logId	int	Unique identifier for the log entry
timestamp	DateTime	When the event occurred
actionType	String	Description of the action performed
targetEmployeeId	int	Foreign key referencing the affected employee
performedBy	String	Username of the actor, or "System" for automated actions
status	String	Outcome of the action (SUCCESS or FAILED)
details	String	Additional context or error messages

Class: Notification

Attribute	Type	Description
notificationId	int	Unique identifier for the notification
recipientId	int	Foreign key referencing the recipient user
taskId	int	Foreign key referencing the related lifecycle task
message	String	The notification message body
type	NotificationType	Severity: INFO, WARNING, or ERROR
isRead	boolean	Whether the recipient has read the notification
createdAt	DateTime	Timestamp when the notification was created

Operation	Signature	Description
markAsRead	markAsRead(): void	Sets isRead to true
send	send(): void	Dispatches the notification to the recipient

Class: Report

Attribute	Type	Description
reportId	int	Unique identifier for the report

Attribute	Type	Description
reportType	ReportType	COMPLIANCE, ACTIVITY_SUMMARY, or FAILED_ACTIONS
startDate	Date	Beginning of the report date range
endDate	Date	End of the report date range
generatedBy	int	Foreign key referencing the user who generated the report
generatedAt	DateTime	Timestamp when the report was generated
departmentFilter	String	Department IDs to filter by (null = all departments)

Operation	Signature	Description
generate	generate(): void	Queries audit logs and task data to populate the report
exportCSV	exportCSV(): File	Exports the report data as a CSV file

Class: SystemConfiguration

Global settings controlling automation behavior.

Attribute	Type	Description
configId	int	Unique identifier for the configuration record
pollingInterval	int	Interval in minutes between OrangeHRM polls
retryLimit	int	Maximum retry attempts for failed operations
notificationsEnabled	boolean	Whether the notification system is active
lastSyncTimestamp	DateTime	Timestamp of the most recent successful sync

Class: ADAccount

An Active Directory user account managed by this system.

Attribute	Type	Description
accountId	int	Unique identifier for the AD account record
employeeId	int	Foreign key referencing the associated employee
samAccountName	String	The AD sAMAccountName (login name)
distinguishedName	String	The full LDAP distinguished name
isEnabled	boolean	Whether the account is currently enabled
createdAt	DateTime	Timestamp when the AD account was created

Attribute	Type	Description
disabledAt	DateTime	Timestamp when the account was disabled

Operation	Signature	Description
enable	enable(): void	Enables the AD account via LDAP
disable	disable(): void	Disables the AD account via LDAP
addToGroup	addToGroup(groupDN: String): void	Adds this account to the specified AD security group
removeFromGroup	removeFromGroup(groupDN: String): void	Removes this account from the specified AD security group

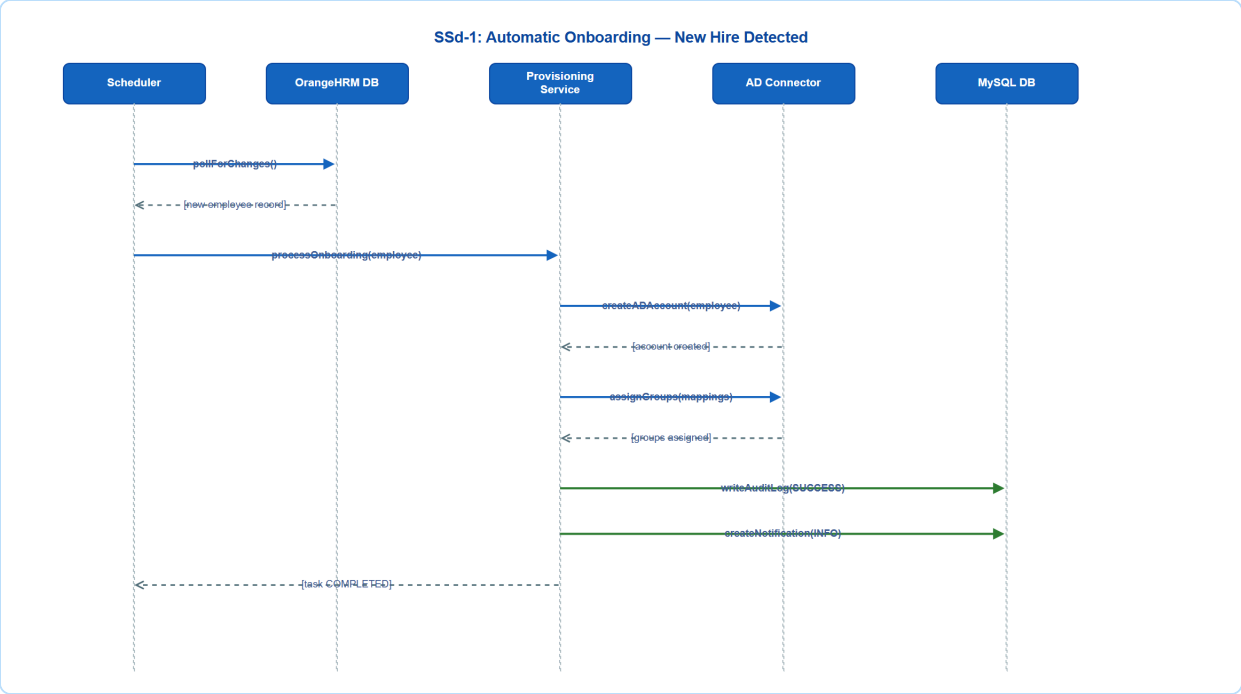
4.7 Class Relationships

Relationship	Type	Multiplicity	Description
User — Department	Association	Many-to-one	Each User belongs to one Department
Employee — Department	Association	Many-to-one	Each Employee belongs to one Department
Employee — ADAccount	Association	One-to-one	Each Employee has at most one ADAccount
Employee — LifecycleTask	Association	One-to-many	Each Employee may have multiple LifecycleTasks
Department — RolePermissionMapping	Association	One-to-many	Each Department may have multiple mappings
RolePermissionMapping — SecurityGroup	Association	Many-to-one	Each mapping references one SecurityGroup
LifecycleTask — TaskAction	Composition	One-to-many	Each task contains one or more TaskActions
LifecycleTask — AuditLog	Association	One-to-many	Each task generates one or more AuditLog entries
LifecycleTask — Notification	Association	One-to-many	Each task may trigger one or more Notifications
User — Notification	Association	One-to-many	Each User may receive multiple Notifications

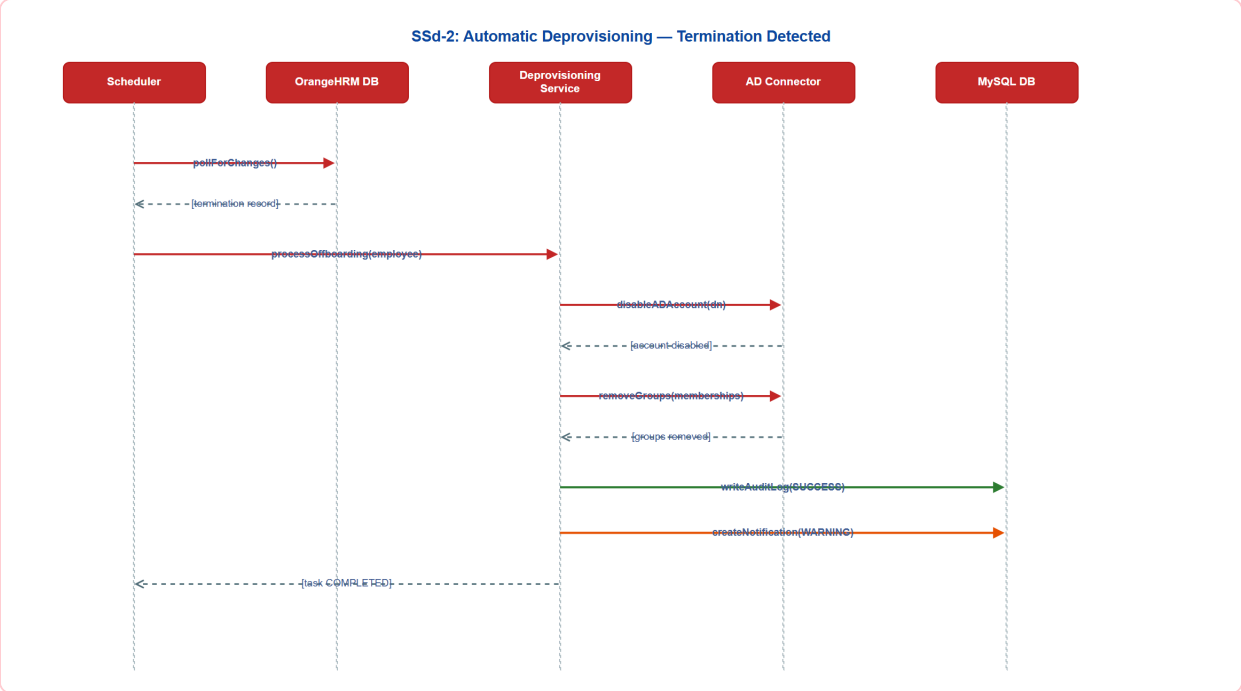
Relationship	Type	Multiplicity	Description
User — Report	Association	One-to-many	Each User may generate multiple Reports

5. System Sequence Diagrams

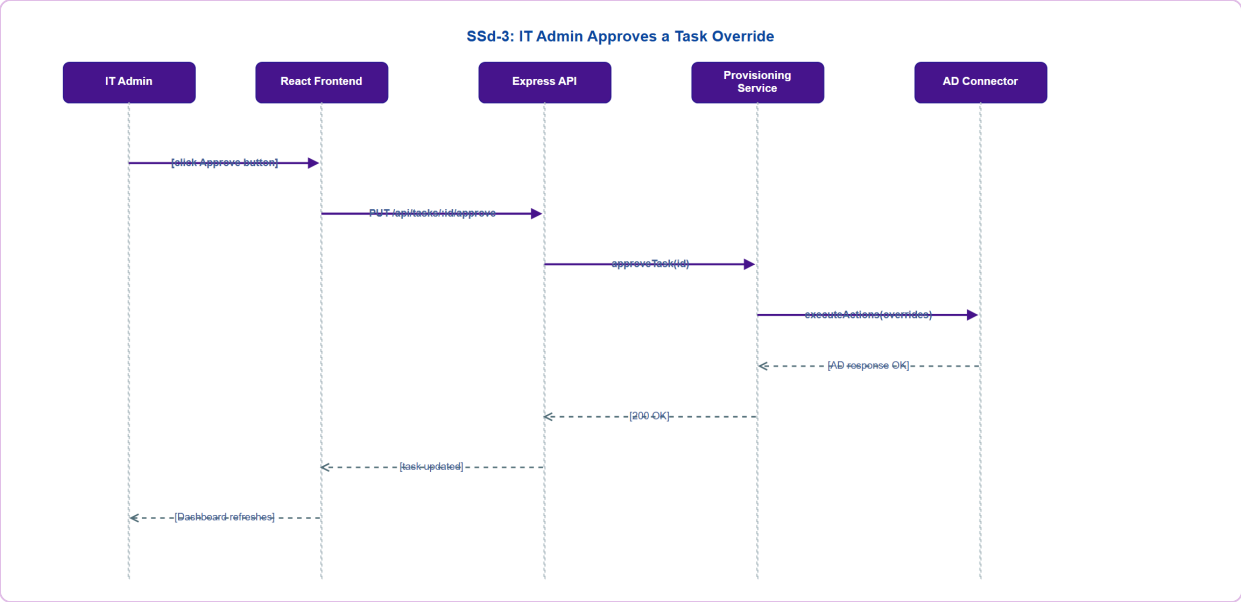
SSd-1: Automatic Onboarding (New Hire Detected)



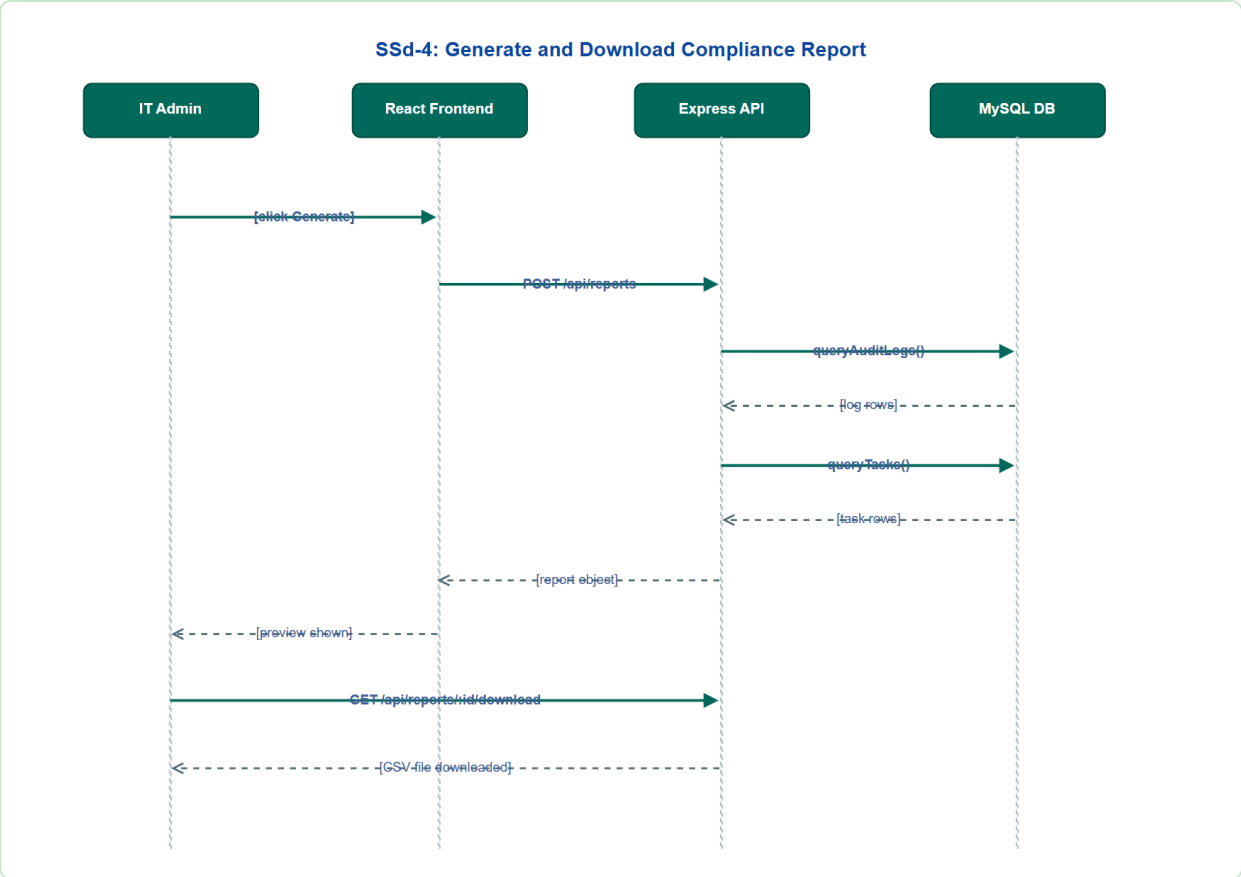
SSd-2: Automatic Deprovisioning (Termination Detected)



SSd-3: IT Admin Approves a Task Override

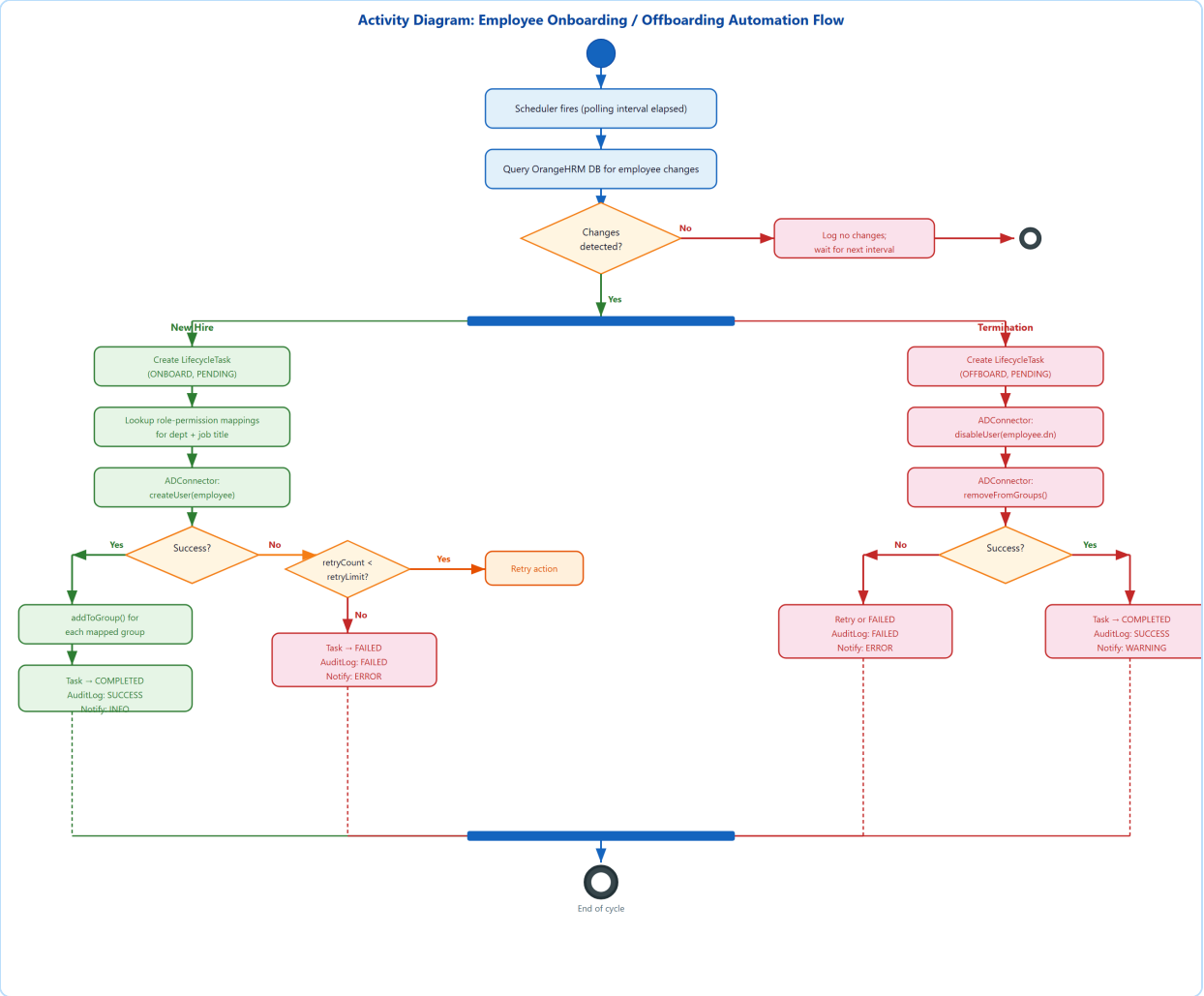


SSd-4: Generate and Download Compliance Report



6. Activity Diagram

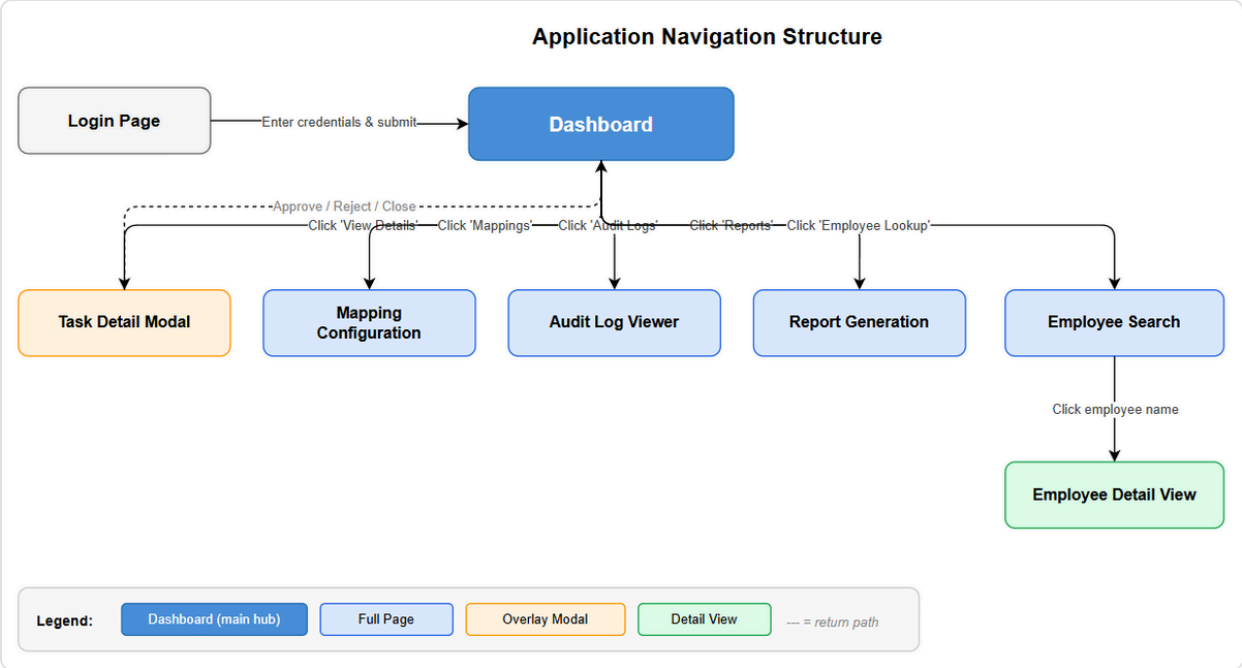
Employee Onboarding/Offboarding Automation Flow



7. User Interface Specification

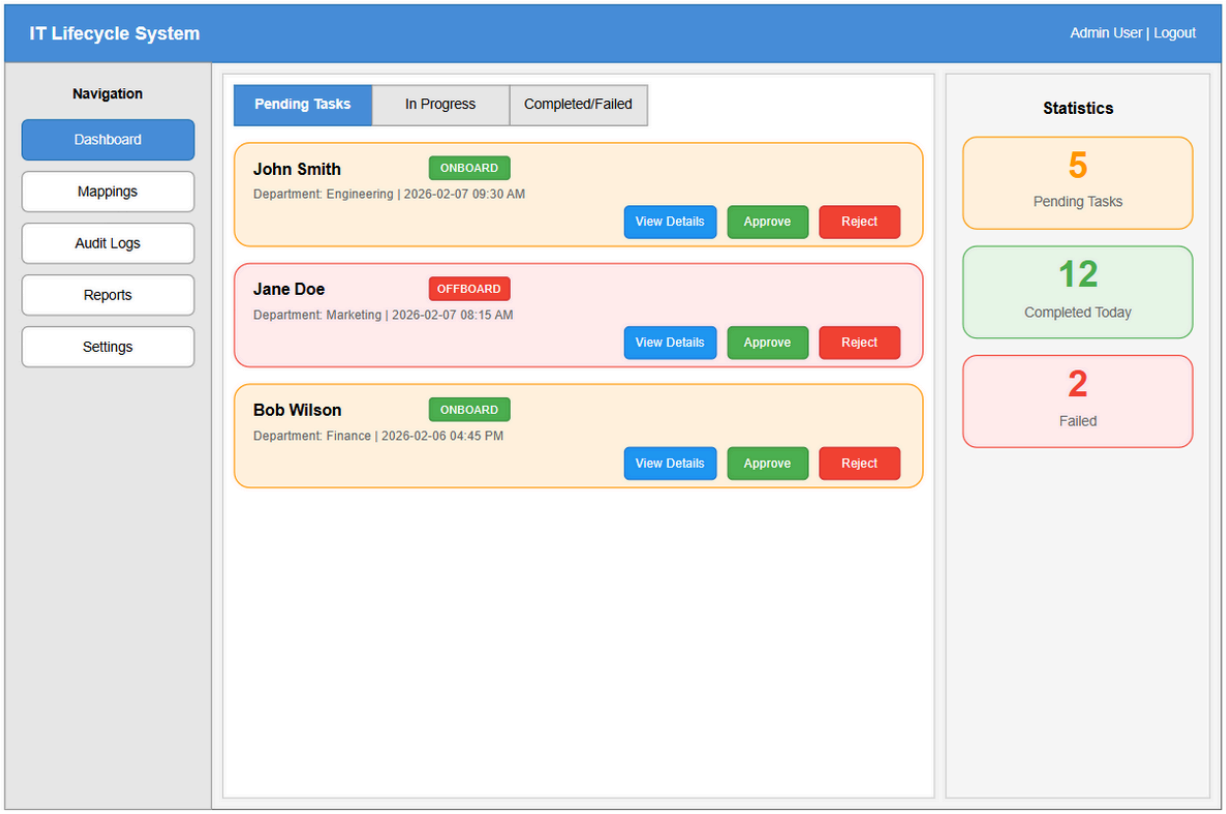
7.1 Global Layout and Navigation

Every screen follows a consistent three-zone layout: a top AppBar with notification bell and user session, a persistent left sidebar with navigation links, and a main content area.

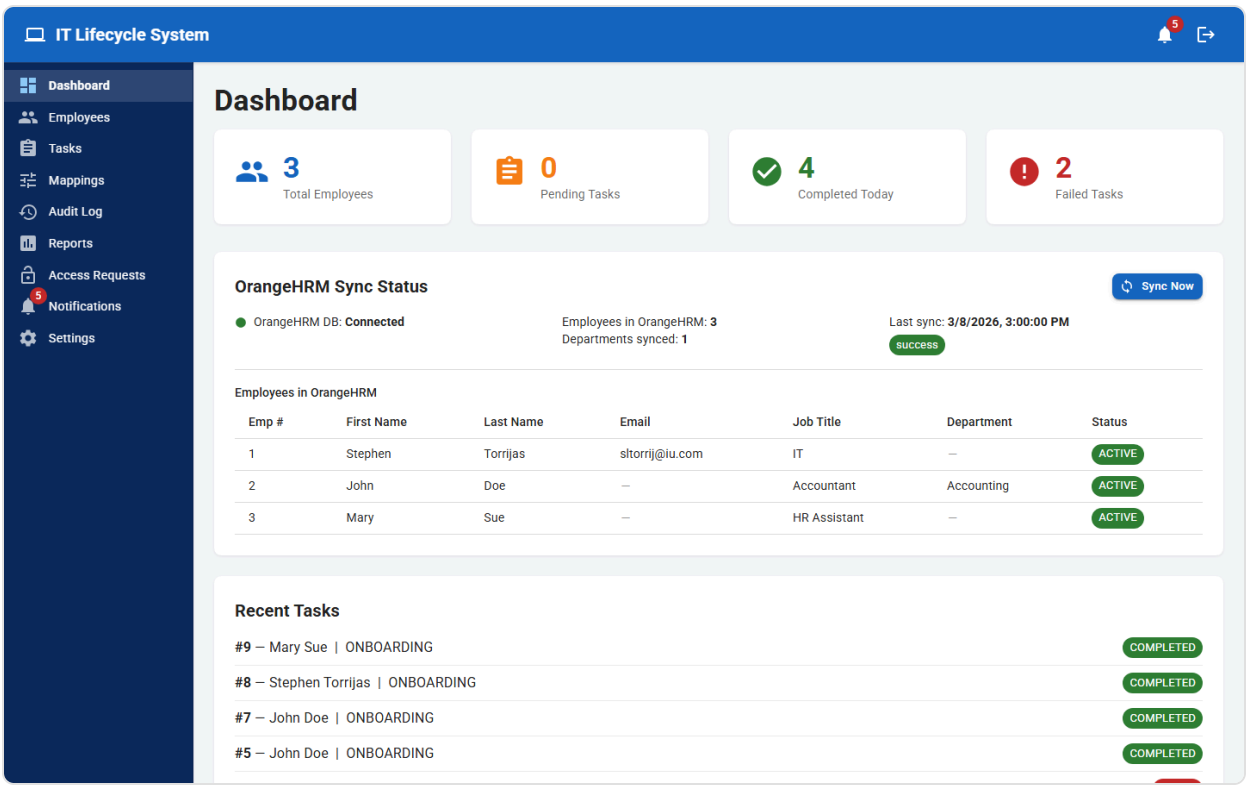


7.2 UI-1: Main Dashboard

Description: The landing page after login. Provides a real-time overview of all lifecycle task activity and OrangeHRM sync status.

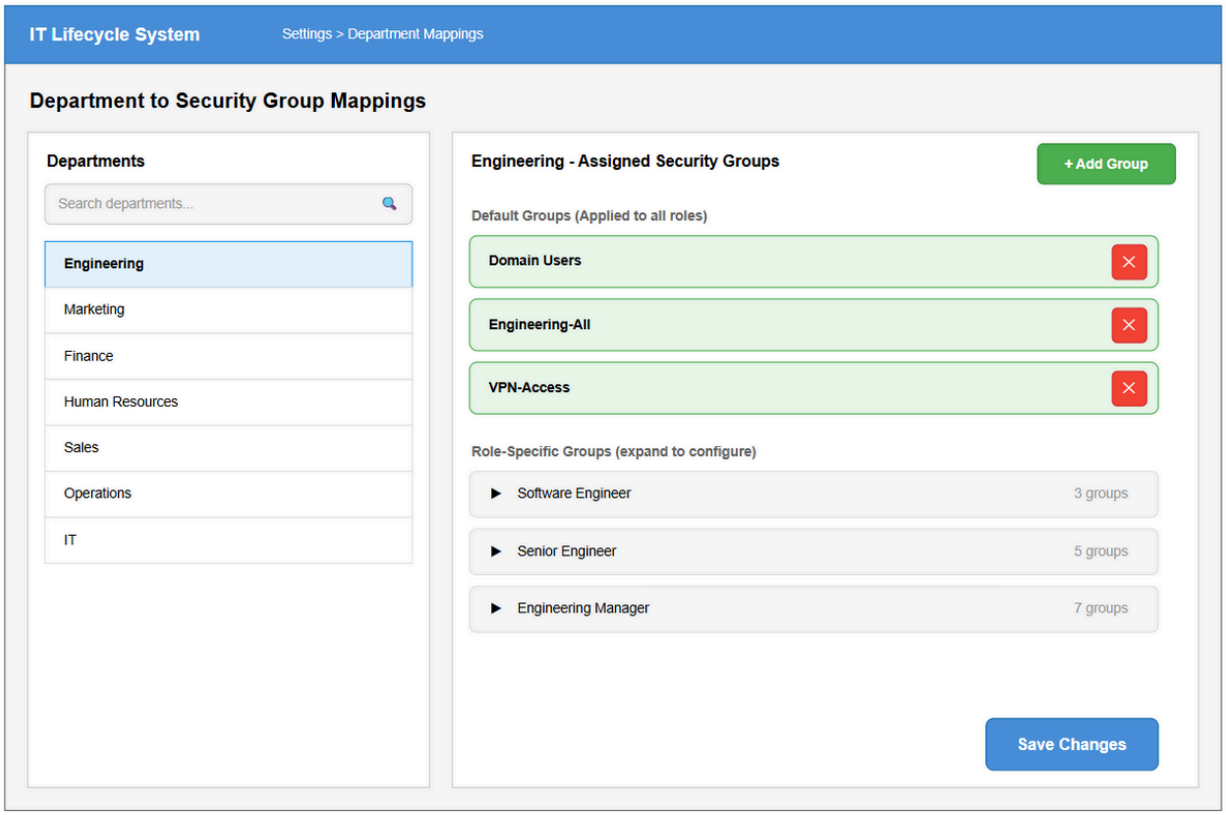


UI-1: Main Dashboard



7.3 UI-2: Department Mapping Configuration

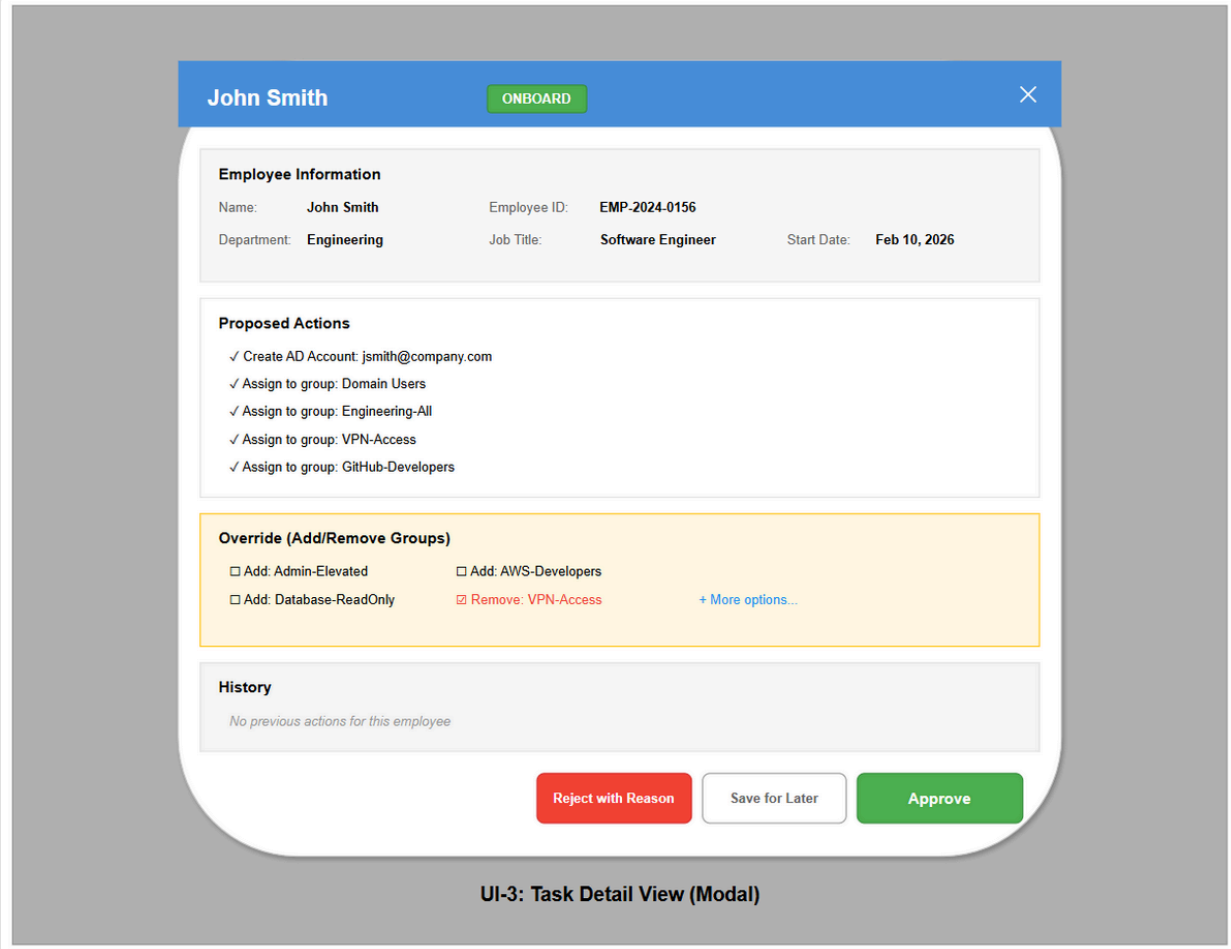
Description: Allows IT Administrators to define which AD security groups are assigned to employees based on department and role.



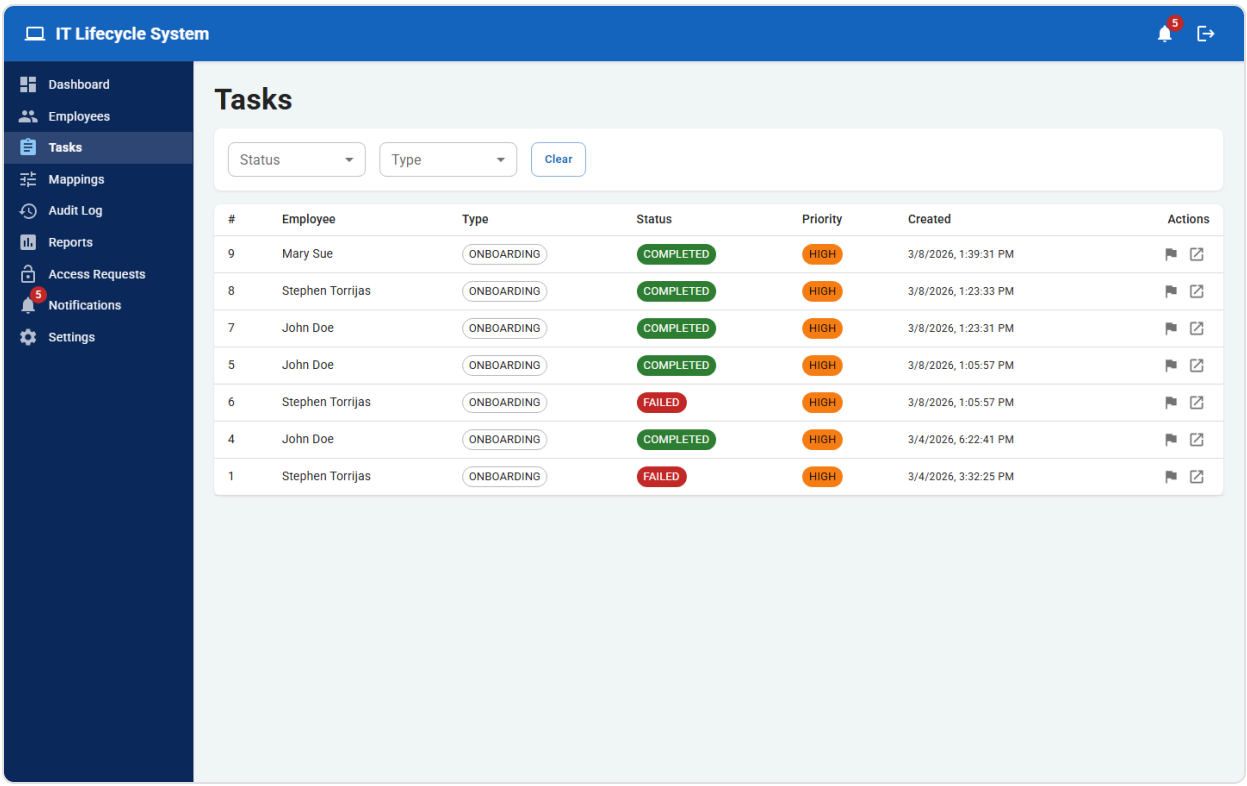
UI-2: Department-to-Group Mapping Configuration

7.4 UI-3: Task Detail View (Modal)

Description: A modal overlay showing full employee details, proposed AD actions, and override options before approval.

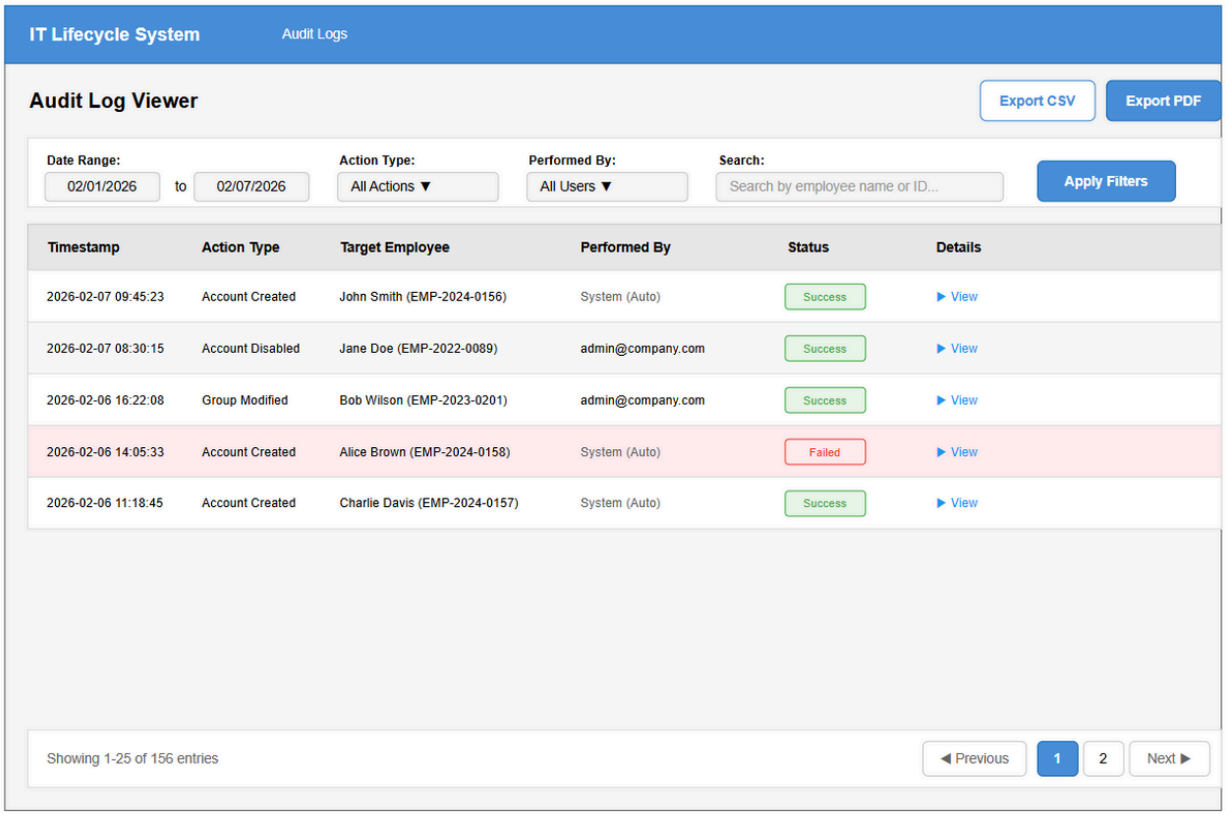


UI-3: Task Detail View (Modal)

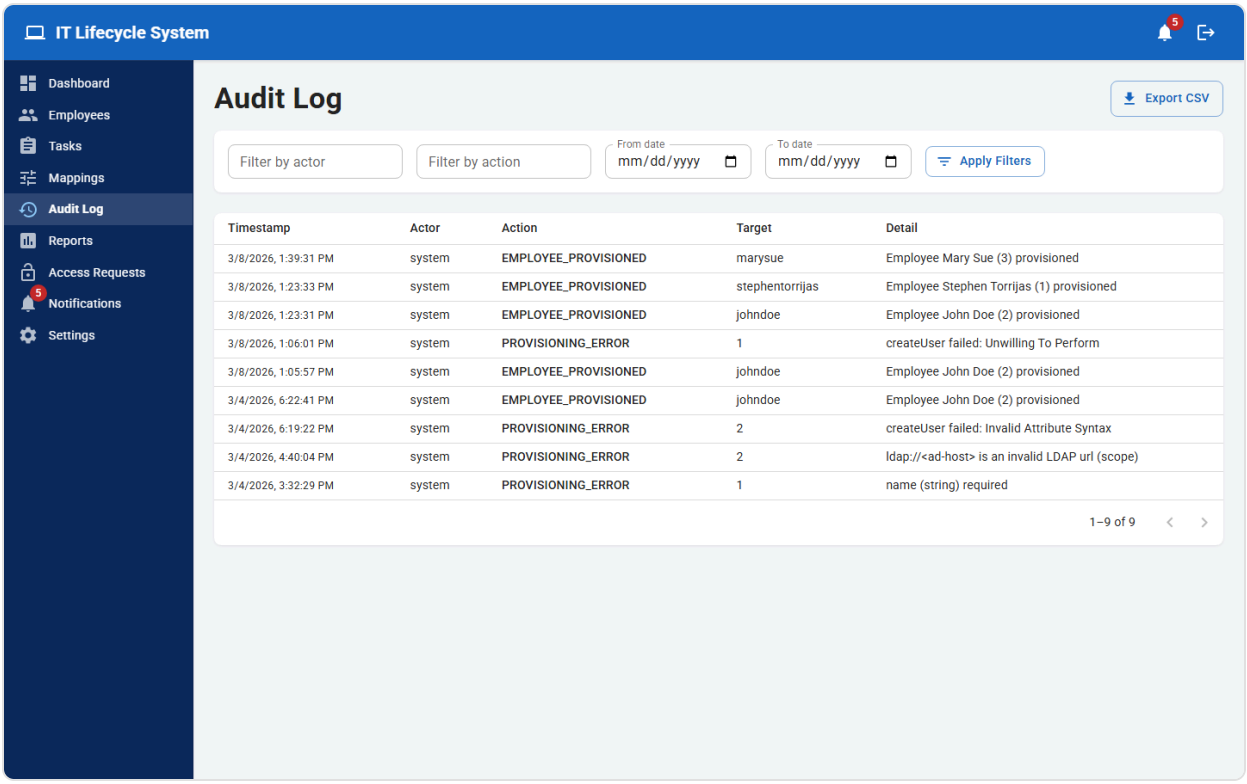


7.5 UI-4: Audit Log Viewer

Description: Searchable, filterable record of all account lifecycle events with CSV export.

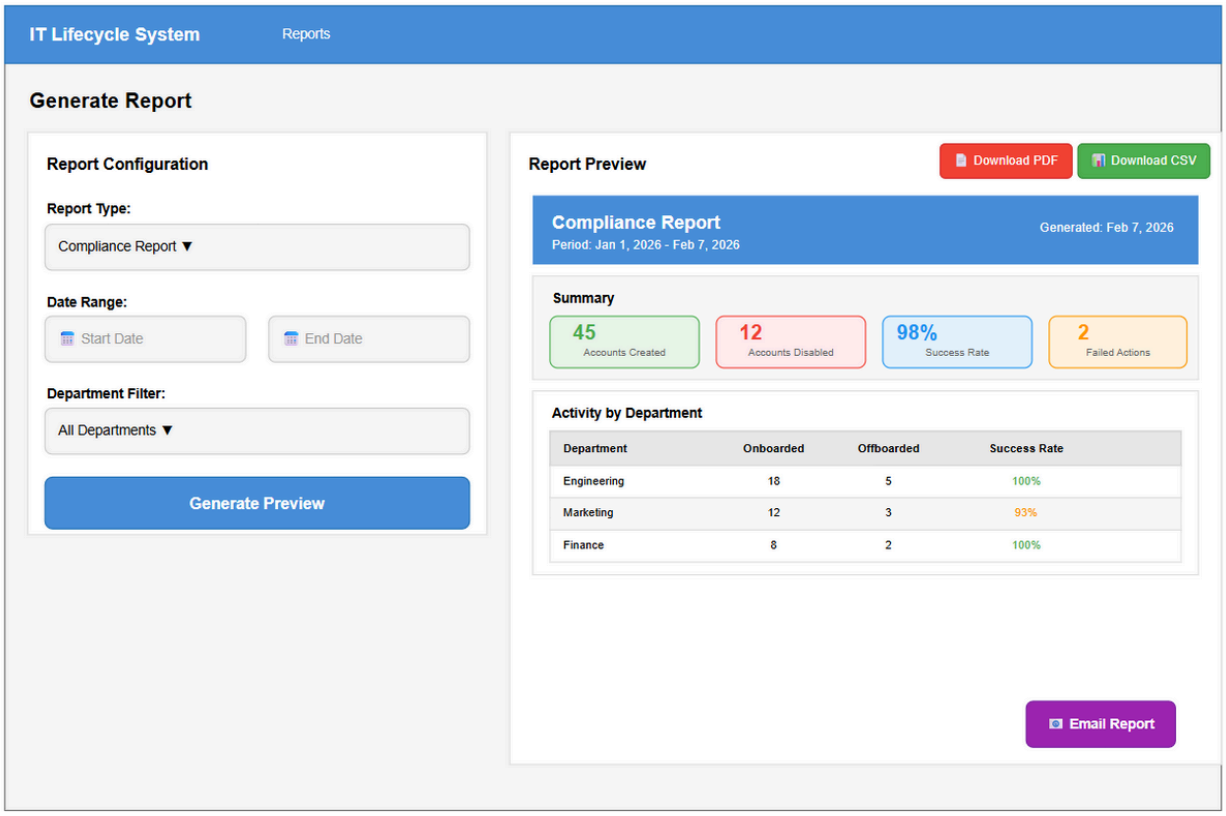


UI-4: Audit Log Viewer



7.6 UI-5: Report Generation

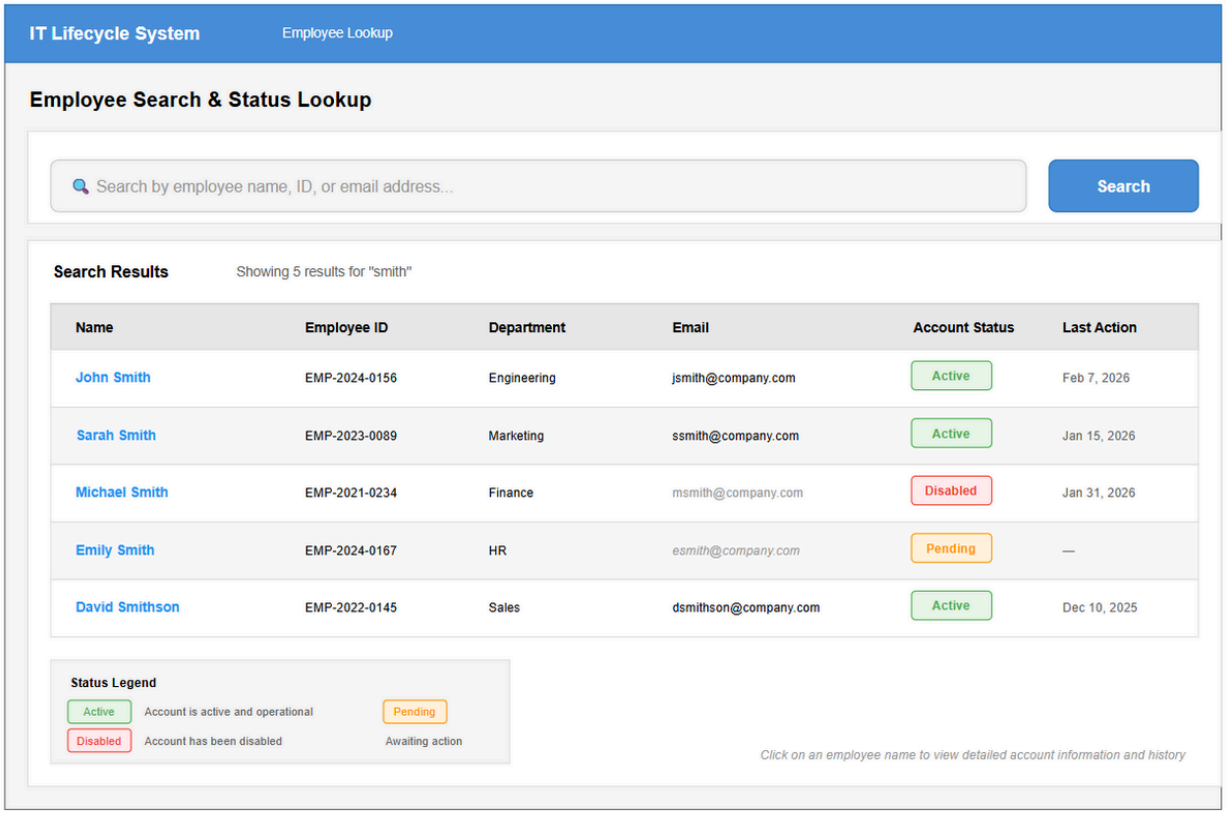
Description: Form-based interface for generating and downloading compliance and activity reports.



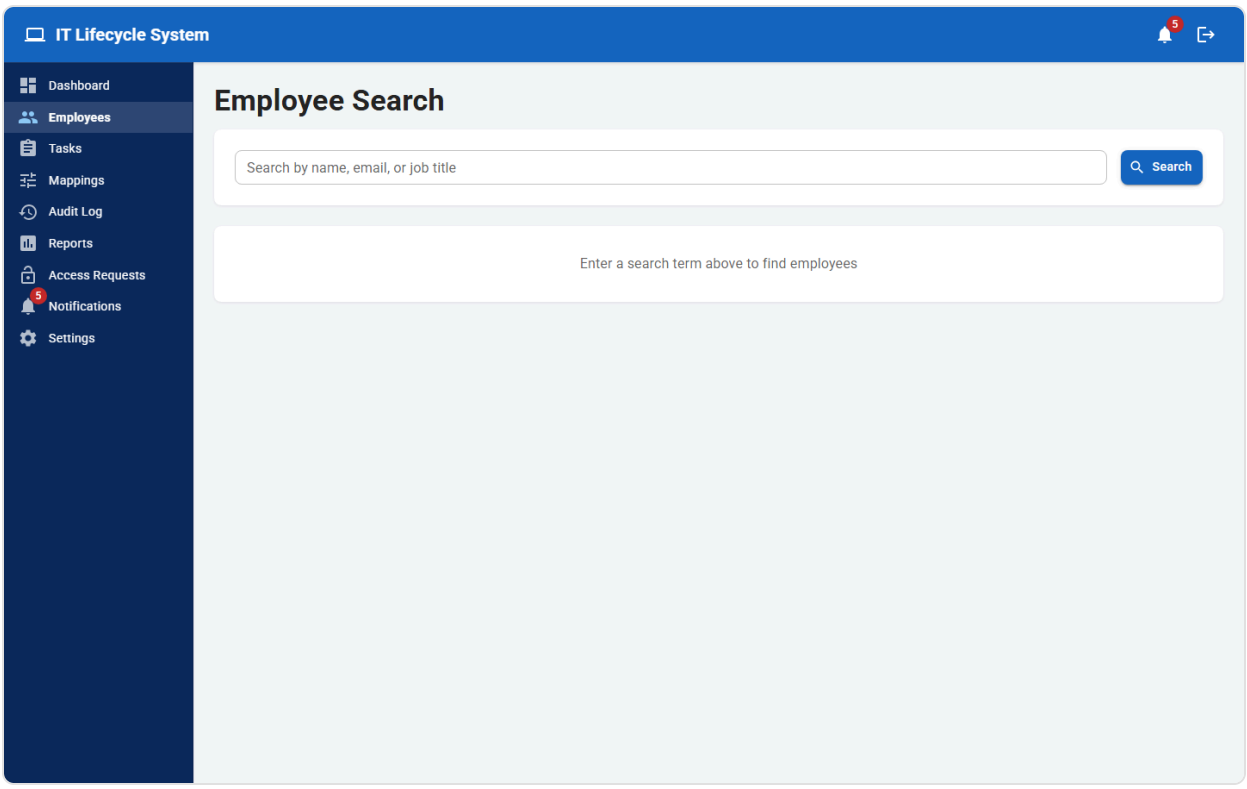
UI-5: Report Generation

7.7 UI-6: Employee Search & Status Lookup

Description: Search employees by name, ID, or email; click a row to view AD account details.



UI-6: Employee Search & Status Lookup



7.8 User Effort Estimation

Scenario	Mouse Clicks	Keystrokes	Total
A: Quick-approve a pending task	2	0	2
B: Approve with group override	6	0	6
C: Search for employee by name	4	5–6	9–10
D: Add a new group mapping	7	7	14
E: Generate and download a report	13	0	13
F: Filter audit logs for failures	5	0	5

The most common operations (approving a task, searching for an employee, viewing audit logs) require 5 or fewer total interactions, consistent with the system's goal of reducing IT workload.

8. Project Plan

Development Schedule

Week	Status	Milestones
1–2	Complete	Project proposal submitted and approved
3–4	Complete	System requirements documentation; environment setup (OrangeHRM + AD VM)
5–7	Complete	MySQL schema deployed; React frontend connected to Node.js backend; core dashboard UI built
8	Complete	Mid-term milestone: core onboarding automation working end-to-end; demo recording
9–11	Complete	Offboarding automation; notification system; audit logging; access request workflow
12–14	In Progress	Reporting and analytics; approval workflow refinements; unit and integration tests
15	Planned	Final integration testing; final demo recording; complete project documentation

Current Implementation Status

All 15 functional requirements (REQ-1 through REQ-15) have been implemented:

REQ	Feature	Status
REQ-1, REQ-4	OrangeHRM polling (new hires + terminations)	Complete
REQ-2	AD account creation via LDAP	Complete
REQ-3	Security group assignment from mappings	Complete
REQ-5	AD account disable on termination	Complete
REQ-6	Role-permission mapping configuration UI	Complete
REQ-7	Task approve/reject/override workflow	Complete
REQ-8	Configurable polling interval	Complete
REQ-9	Audit log with date filter and CSV export	Complete
REQ-10	Compliance report generation + CSV download	Complete

REQ	Feature	Status
REQ-11, REQ-13	Employee search with AD account detail	Complete
REQ-12	High-priority task flagging	Complete
REQ-14	Access request submission and approval	Complete
REQ-15	In-app notifications with unread badge	Complete

Planned for Final Submission

- LDAPS (encrypted AD communication on port 636)
- Role-based UI views (HR Personnel, Department Manager scoped views)
- Unit tests (Jest) for provisioning, deprovisioning, and notification services
- Integration tests for end-to-end lifecycle flows

Software Stack

Layer	Technology
Frontend	React 18, Material UI (MUI)
Backend	Node.js, Express 4
Database	MySQL 8 (12-table schema)
HR Integration	OrangeHRM 5.x (direct MySQL polling via node-schedule)
Directory	Windows Active Directory, ldapjs (LDAP)
Auth	JWT (jsonwebtoken)

9. References

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